# AI and LLM Powered Personalized Lifestyle Risk Awareness System (Web App)

*A Review Report Submitted in Partial Fulfillment of the Requirement for the Degree*

*of*

**Master of Technology**

**in**

**Computer Science and Engineering**
**(AI & DS)**

*Submitted by*

**Anushka Khatua**
***Roll No. - 24167030***

*Under the Guidance*
*of*

**Prof. Bapuji Kanaparthi**

**School of Computer Engineering,**
**Kalinga Institute of Industrial Technology University**
**Deemed to be University**
**Bhubaneswar, Odisha, India**
**Session:-2025-2026**

**School of Computer Engineering**

**Kalinga Institute of Industrial Technology**

**(Deemed to be University), Bhubaneswar**

**Odisha, India.**

# CERTIFICATE

This is to certify that this review report entitled ***"AI and LLM Powered Personalized Lifestyle Risk Awareness System (Web App)"***, submitted by ***Anushka Khatua***, bearing ***Roll No: 24167030*** in the partial fulfillment of the requirement for the degree of ***Master of Technology in Computer Science and Engineering (AI & DS)*** at School of Computer Engineering, Kalinga Institute of Industrial Technology(Deemed to be University), Bhubaneswar, has been carried out under my supervision. To the best of my knowledge the matter embodied in the report has not been submitted to any other university/institute for the award of any degree or diploma.

**(Prof. Bapuji Kanaparthi)**

**Supervisor**

**School of Computer Engineering**

**KIIT University, Bhubaneswar**

**Odisha, India.**

# DECLARATION

I do hereby declare that this report entitled ***"AI and LLM Powered Personalized Lifestyle Risk Awareness System (Web App)"***, submitted in partial fulfillment of the requirement for the award of degree of Master of Technology in Computer Science and Engineering (AI & DS) at School of Computer Engineering, Kalinga Institute of Industrial Technology (KIIT), Deemed to be University, Bhubaneswar is my own and has not been submitted to any other institute.

**Anushka Khatua**

**Roll No.- 24167030**

**M.Tech.(Computer Science and Engineering(AI & DS))**

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**Anushka Khatua**

**Roll No.- 24167030**

**M.Tech.(Computer Science and Engineering(AI & DS))**

# Abstract

Modern lifestyles are increasingly associated with unhealthy habits such as inadequate sleep, low physical activity, excessive screen time, and irregular daily routines, which can lead to long-term health risks. Addressing the lack of personalized awareness in existing wellness systems, this project presents an AI and LLM-powered Personalized Lifestyle Risk Awareness System developed as a web-based application. The system analyzes user lifestyle data, including activity patterns and behavioral indicators, to assess overall lifestyle risk. Machine learning and deep learning models are utilized to classify users into different risk levels such as Low, Medium, and High. The predicted outcomes are further processed using a Large Language Model (LLM) to generate clear, user-friendly insights and recommendations, enhancing interpretability and user understanding. The application integrates multiple modules, including data preprocessing, exploratory data analysis, model development, and a web interface for user interaction. It is implemented using Python-based technologies with a backend API for model integration and deployed as an interactive platform. This system serves as a practical tool for promoting lifestyle awareness by combining predictive analytics with explainable insights, enabling users to better understand their habits and make informed decisions toward healthier living.

**Keywords: Artificial Intelligence, Lifestyle Risk Assessment, Large Language Models (LLM), Health Analytics, Web Application, Predictive Modeling, Human Activity Recognition**

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# Chapter 1

# Introduction

## 1.1 Background

In recent years, modern lifestyles have undergone significant changes due to increased use of technology and sedentary routines. Factors such as reduced physical activity, irregular sleep patterns, and excessive screen time have become common, contributing to various health risks including stress, fatigue, and long-term diseases. Although many digital health applications exist, most provide generic recommendations and do not consider individual lifestyle differences. There is a growing need for systems that can analyze personal behavior data and provide customized insights. Advancements in Artificial Intelligence (AI), Machine Learning (ML), and Large Language Models (LLMs) enable effective analysis of lifestyle patterns and generation of meaningful insights. This project utilizes these technologies to develop a system that assesses lifestyle risks and promotes awareness through personalized, data-driven feedback.

## 1.2 Problem Statement

Modern lifestyle habits, including insufficient physical activity, poor sleep quality, excessive screen time, and unhealthy daily routines, are increasingly contributing to long-term health risks. Existing digital health solutions primarily focus on clinical diagnosis or provide generic wellness advice, lacking personalization and proactive risk awareness.

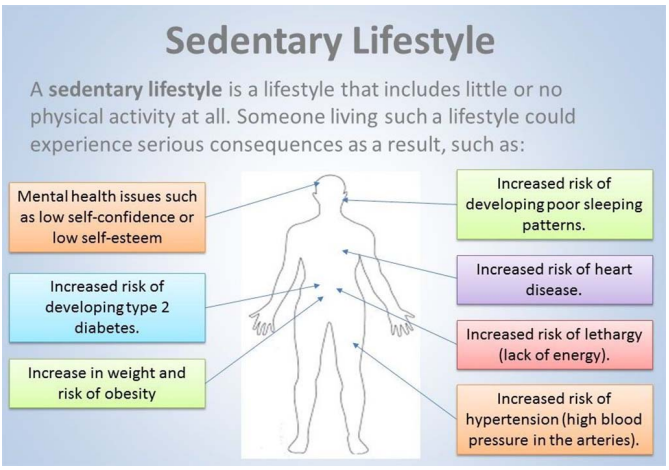


Figure 1.1: Sedentary Lifestyle Side Effects

There is a need for an intelligent system that can analyze individual lifestyle data, identify behavioral patterns, and provide personalized risk assessment along with clear and understandable insights. Such a system should bridge the gap between raw data analysis and user-friendly interpretation, enabling individuals to become more aware of their lifestyle choices and their potential long-term impact.

## 1.3 Objective

- To develop a system that analyzes lifestyle behavior data for risk assessment
- To implement Machine Learning and Deep Learning models for predicting lifestyle risk levels
- To classify users into different risk categories such as Low, Medium, and High
- To integrate a Large Language Model (LLM) for generating personalized and interpretable insights
- To design and develop an interactive web-based application for user input and visualization
- To provide data-driven, actionable recommendations for improving lifestyle habits
- To ensure the system is scalable, user-friendly, and suitable for real-time analysis

# Chapter 2

# Literature Review

Table 2.1: Comparative Analysis of Existing Methods

| Ref No | Paper | Model Used | Performance | Key Findings |
|---|---|---|---|---|
| [7] | Nouriani et al. (2022) | Deep Neural Networks | High accuracy (dataset dependent) | Demonstrated effectiveness of deep learning in capturing complex temporal patterns from wearable sensor data for activity recognition. |
| [8] | Qin & Tong (2025) | Large Language Models (LLMs) | Not accuracy-based | Highlighted the role of LLMs in generating personalized healthcare insights and improving patient interaction, with challenges in reliability. |

*Continued on next page*

| Ref No | Paper | Model Used | Performance | Key Findings |
|---|---|---|---|---|
| [3] | Bulling et al. (2014) | Activity Recognition Chain (ARC), ML models | Framework-based | Proposed a complete HAR pipeline including preprocessing, feature extraction, and classification, and discussed challenges like variability and noise. |
| [4] | Chen et al. (2021) | CNN, RNN, LSTM | Up to 95%+ | Deep learning models automatically extract features and outperform traditional ML methods in activity recognition tasks. |
| [1] | Anguita et al. (2013) | Multiclass SVM | 96% accuracy | Demonstrated that smartphone-based sensors can effectively perform activity recognition with high accuracy. |
| [6] | Luwe et al. (2022) | 1D-CNN + BiLSTM | 95.48%–100% | Hybrid deep learning models improve performance by combining spatial and temporal feature learning. |

*Continued on next page*

| Ref No | Paper | Model Used | Performance | Key Findings |
|---|---|---|---|---|
| [5] | Lara & Labrador (2013) | ML (SVM, Decision Trees, etc.) | 92%–98% | Wearable sensor-based HAR systems achieve high accuracy but depend heavily on feature engineering and system design. |

# Chapter 3

# Methodology

## 3.1 Overview of the System

The proposed system is an AI-driven web-based platform designed to analyze lifestyle behavior and assess associated risk levels. It integrates Machine Learning (ML), Deep Learning (DL), and Large Language Model (LLM) components to provide both predictive and interpretable outputs. The methodology follows a modular pipeline where user input data is processed, analyzed using trained models, and converted into meaningful insights.

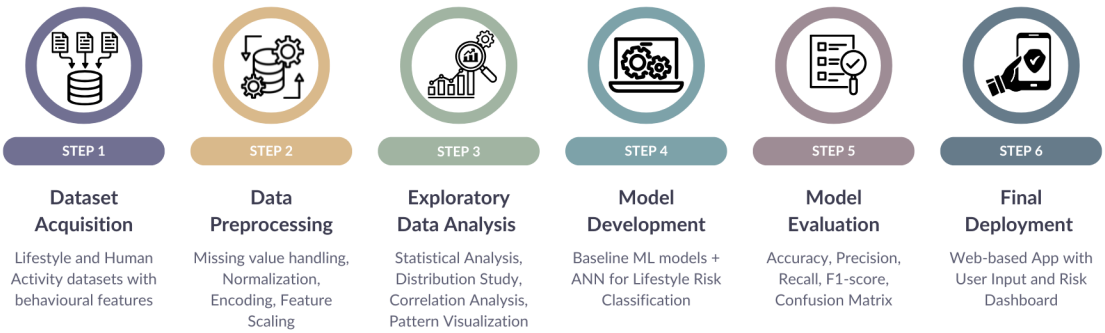


Figure 3.1: Methodology Flowchart

## 3.2 Data Acquisition

The system utilizes multiple publicly available datasets containing lifestyle and activity-related information. These datasets include behavioral patterns such as physical activity, daily routines, and movement signals. Data is collected from sources like Human Activity Recognition datasets [10], fitness tracking data [2], and daily activity datasets [9]. Combining multiple datasets improves diversity, coverage, and model robustness by incorporating different lifestyle patterns.

## 3.3 Data Preprocessing

Raw data often contains inconsistencies, missing values, and variations in scale. Therefore, preprocessing is performed to ensure data quality and consistency.

- Key preprocessing steps include:
- Handling missing or inconsistent values
- Feature normalization to a standard range
- Data cleaning and formatting
- Feature alignment across multiple datasets

This step ensures that the data is suitable for training machine learning and deep learning models.

## 3.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to understand patterns and relationships within the dataset. It helps in identifying trends in physical activity, sedentary behavior, and lifestyle habits. Visualization techniques such as distribution plots and activity comparisons are used to analyze data characteristics. EDA also helps in detecting outliers and ensuring balanced class distribution for effective model training.

## 3.5 Feature Engineering

Feature engineering is carried out to extract meaningful information from raw data. Important features related to lifestyle habits, activity levels, and behavioral patterns are selected and transformed. This process improves model performance by reducing noise and highlighting relevant attributes that contribute to accurate risk prediction.

## 3.6 Model Development

The system employs multiple models for lifestyle risk prediction:

- Logistic Regression for baseline classification
- Random Forest for handling complex feature interactions
- Artificial Neural Network (ANN) for capturing non-linear relationships

These models are trained to classify users into risk levels such as Low, Medium, and High based on their lifestyle data.

## 3.7 Model Evaluation

The performance of the models is evaluated using standard metrics such as:

- Accuracy
- Precision
- Recall
- F1-score

A confusion matrix is also used to analyze classification performance. This step ensures that the selected model provides reliable and consistent predictions.

## 3.8 LLM-Based Insight Generation

To enhance interpretability, the system integrates a Large Language Model (LLM). The LLM converts model predictions into human-readable explanations and personalized recommendations. This component bridges the gap between technical outputs and user understanding, making the system more practical and user-friendly.

## 3.9 System Integration

All components are integrated into a unified pipeline:

- User input is collected through the web interface
- Data is processed and passed to the trained model
- Predictions are generated and sent to the LLM module
- Final insights are displayed to the user

This ensures smooth data flow and real-time processing.

## 3.10 Deployment Strategy

The system is deployed as a web application using backend frameworks such as Flask or FastAPI. The trained models and LLM are integrated into the backend, while the frontend provides an interactive interface for users. Deployment on platforms such as Hugging Face enables accessibility, scalability, and real-time interaction.

# Chapter 4

# System Architecture

## 4.1 Overview

The system architecture of the proposed Lifestyle Risk Awareness System is designed as a modular and scalable pipeline that integrates user interaction, data processing, machine learning models, and insight generation. The architecture ensures smooth data flow from user input to final output, enabling real-time analysis and personalized feedback. The system follows a layered approach consisting of frontend, backend, model processing, and insight generation components.

## 4.2 Components of the System

### 4.2.1 User Interface (Frontend)

The frontend is responsible for collecting user input related to lifestyle behaviors such as activity levels, daily routines, and habits. It provides an interactive and user-friendly interface where users can enter their data and view results. It also displays outputs in the form of:

- Risk score
- Risk category (Low / Medium / High)

- Visual analytics (charts and graphs)
- Personalized insights

### 4.2.2 Backend API

The backend acts as the core processing unit of the system. It is implemented using frameworks such as Flask or FastAPI. Its main functions include:

- Receiving user input data from the frontend
- Preprocessing and formatting the data
- Sending data to the trained ML/DL model
- Managing communication between different modules

### 4.2.3 Machine Learning / Deep Learning Module

This module is responsible for predicting lifestyle risk levels. It consists of trained models such as Logistic Regression, Random Forest, and Artificial Neural Network. The module:

- Takes processed input data
- Performs classification
- Outputs a risk level (Low / Medium / High) along with a risk score

### 4.2.4 LLM Insight Generation Module

The LLM module enhances interpretability by converting model outputs into human-readable explanations. It includes:

- Interprets prediction results
- Generates personalized feedback
- Provides actionable recommendations

This makes the system more user-centric and understandable.

### 4.2.5 Visualization and Dashboard Module

This module presents the results in an intuitive format using charts and graphical representations. It helps users understand their lifestyle patterns and risk distribution effectively.

## 4.3 Data Flow Architecture

- User enters lifestyle data through the frontend
- Data is sent to the backend API
- Backend preprocesses the data
- Processed data is passed to the ML/DL model
- Model predicts the risk level
- Output is sent to the LLM module
- LLM generates insights and recommendations
- Final results are displayed on the dashboard

## 4.4 System Characteristics

- Modular Design – Each component functions independently
- Scalability – Can handle multiple users and datasets
- Real-time Processing – Provides instant predictions
- User-friendly Interface – Easy interaction and understanding
- Integration Capability – Supports ML, DL, and LLM together

# Chapter 5

# Model Development

## 5.1 Overview

This chapter focuses on the development of machine learning and deep learning models for lifestyle risk prediction. The objective is to classify users into different risk levels based on their lifestyle behavior data. Multiple models are implemented and compared to ensure accurate and reliable prediction.

## 5.2 Machine Learning Models

### 5.2.1 Logistic Regression

Logistic Regression is used as a primary classification model due to its simplicity and effectiveness. It models the probability of a user belonging to a particular risk category. It performs well on structured and normalized data and provides good generalization.

### 5.2.2 Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees and combines their outputs. It helps in handling complex feature interactions and improves prediction accuracy by reducing overfitting.

## 5.3 Deep Learning Model

### 5.3.1 Artificial Neural Network (ANN)

An Artificial Neural Network is implemented to capture non-linear relationships in the data. The architecture consists of:

- Input Layer: Takes lifestyle-related features
- Hidden Layers: Fully connected dense layers with ReLU activation
- Regularization: Dropout layers to prevent overfitting
- Output Layer: Softmax activation for multi-class classification (Low, Medium, High)

The ANN model is trained using backpropagation to optimize performance.

## 5.4 Model Training

The models are trained on preprocessed and normalized data. The dataset is divided into training and testing sets to evaluate performance.

Key aspects of training include:

- Loss Function: Categorical Cross-Entropy
- Optimizer: Adam Optimizer
- Batch Processing for efficient training

## 5.5 Feature Utilization

The models utilize features derived from lifestyle behavior, including physical activity patterns, daily routines, and movement-based signals. Proper feature selection and transformation improve model efficiency and accuracy.

# Chapter 6

# Model Evaluation and Results

## 6.1 Overview

This chapter presents the evaluation of the developed machine learning and deep learning models for lifestyle risk prediction. The performance of each model is analyzed using standard evaluation metrics to determine their effectiveness and reliability.

## 6.2 Evaluation Metrics

To assess model performance, the following metrics are used:

- **Accuracy**: Measures the overall correctness of predictions
- **Precision**: Indicates the proportion of correct positive predictions
- **Recall**: Measures the ability of the model to identify all relevant instances
- **F1-Score**: Harmonic mean of precision and recall

These metrics provide a comprehensive understanding of model performance.

## 6.3 Model Performance Comparison

The performance of the implemented models is summarized as follows:

- Logistic Regression: 96.16% accuracy
- Random Forest: 92.67% accuracy
- Artificial Neural Network: 91.48% accuracy

Among these models, Logistic Regression demonstrates the highest accuracy and better generalization performance.

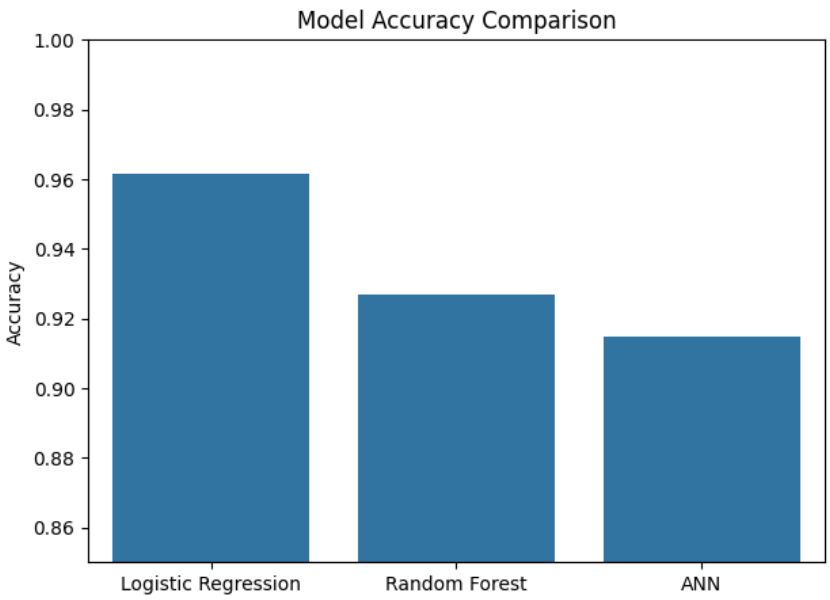


Figure 6.1: Accuracy Comparison

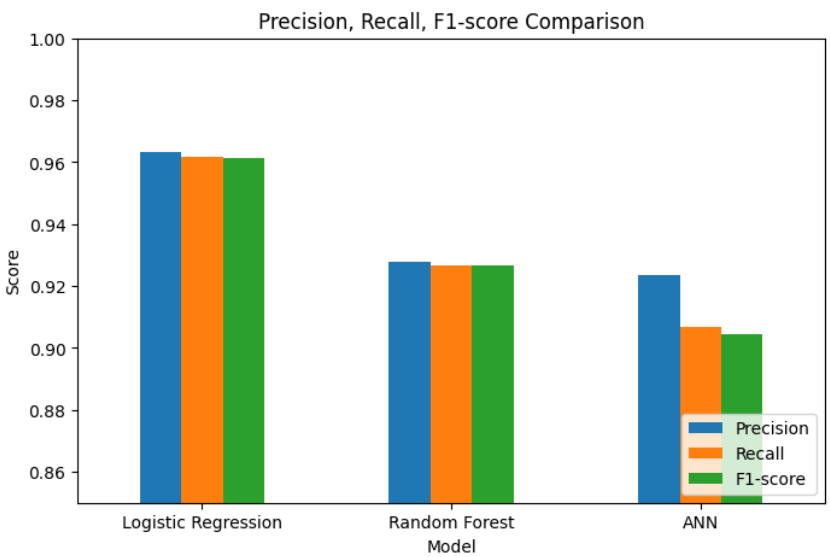


Figure 6.2: Evaluation Metrics Comparison

## 6.4 Confusion Matrix Analysis

A confusion matrix is used to visualize the classification performance of the models. It shows the number of correct and incorrect predictions for each class.

The analysis indicates that the model is able to effectively distinguish between different lifestyle risk categories (Low, Medium, High), with minimal misclassification.

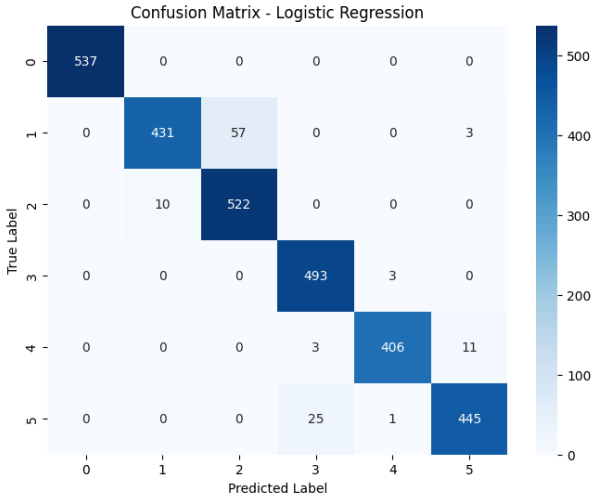


Figure 6.3: Confusion Matrix for Logistic Regression

## 6.5 Result Interpretation

The results indicate that the models are capable of accurately predicting lifestyle risk levels based on behavioral data. The high accuracy values suggest that the system can reliably classify users into appropriate risk categories.

The consistency in performance across different datasets further demonstrates the robustness of the proposed approach.

## 6.6 Summary

Based on the evaluation, Logistic Regression is selected as the most suitable model for deployment due to its superior performance and generalization capability.

# Chapter 7

# LLM Integration

## 7.1 Overview

This chapter describes the integration of a Large Language Model (LLM) into the system to enhance interpretability and user experience. While machine learning models provide numerical predictions, the LLM converts these outputs into meaningful and human-readable insights.

## 7.2 Need for LLM Integration

The predictions generated by ML/DL models are often difficult for users to interpret directly. Risk categories such as Low, Medium, and High do not provide sufficient explanation about the underlying lifestyle patterns.

To address this limitation, an LLM is integrated to:

- Explain prediction results in simple language
- Provide personalized feedback based on user input
- Suggest actionable lifestyle improvements

This improves the usability and effectiveness of the system.

## 7.3 Working Mechanism

The LLM module operates after the prediction stage. The workflow is as follows:

1. The ML/DL model generates a risk prediction
2. Prediction output and relevant features are passed to the LLM
3. Rule-based logic structures the input for interpretation
4. The LLM generates a detailed explanation and recommendations

This ensures that the output is both accurate and user-friendly.

## 7.4 Rule-Based Interpretation Layer

Before passing data to the LLM, a rule-based layer is used to organize prediction results. This includes predefined conditions that map model outputs to meaningful contexts.

For example:

- High risk may be associated with low activity and poor sleep
- Medium risk may indicate moderate activity but inconsistent habits

This structured input improves the quality and relevance of LLM-generated responses.

## 7.5 Personalized Insight Generation

The LLM generates personalized insights based on:

- User lifestyle patterns
- Predicted risk category
- Behavioral trends identified in data

The output includes:

- Explanation of current lifestyle condition
- Identification of risk factors
- Suggestions for improvement

## 7.6 Advantages of LLM Integration

- Enhances interpretability of model predictions
- Improves user engagement and understanding
- Provides personalized and actionable insights
- Bridges the gap between technical output and user comprehension

## 7.7 Summary

The integration of the LLM transforms the system from a simple prediction tool into an intelligent assistant that provides meaningful and personalized lifestyle guidance. This significantly increases the practical value of the application.

# Chapter 8

# Deployment and Implementation

## 8.1 Overview

This chapter describes the implementation and deployment of the proposed AI-based lifestyle risk awareness system. The application is developed as a web-based platform that integrates machine learning models and a Large Language Model (LLM) to provide real-time predictions and personalized insights. The system is designed to be user-friendly, scalable, and accessible through a web interface.

## 8.2 System Implementation

The system is implemented using Python-based technologies for efficient backend processing and model integration. The key components include:

- **Programming Language:** Python
- **Backend Framework:** Flask / FastAPI
- **Machine Learning Libraries:** Scikit-learn, TensorFlow/Keras
- **Data Processing:** NumPy, Pandas
- **Visualization:** Matplotlib, Plotly

The backend processes user inputs, performs data preprocessing, executes trained models, and generates prediction outputs.

## 8.3 Frontend Development

The frontend is designed to provide a clean and interactive interface for users to input their lifestyle data and view results. It is developed using HTML, CSS, and JavaScript. The frontend functionalities include:

- Structured input forms for collecting user data
- Display of health score and risk levels
- Interactive visualization of health metrics
- Presentation of personalized insights and recommendations

## 8.4 Application Interface

The graphical user interface of the system is shown in Figure 8.1 and Figure 8.2. The application is divided into input and output sections to ensure smooth interaction and clear presentation of results.

### 8.4.1 User Input Interface

The input interface is designed to collect all necessary lifestyle-related data from the user. It is divided into multiple sections for better organization:

- **Personal Details:** Age, Gender
- **Health Metrics:** Daily steps, calorie intake, heart rate
- **Lifestyle Factors:** Sleep hours, water intake, activity level, diet type, smoking, and alcohol consumption

The layout is structured and intuitive, allowing users to easily enter their data. A dedicated action button initiates the analysis process.

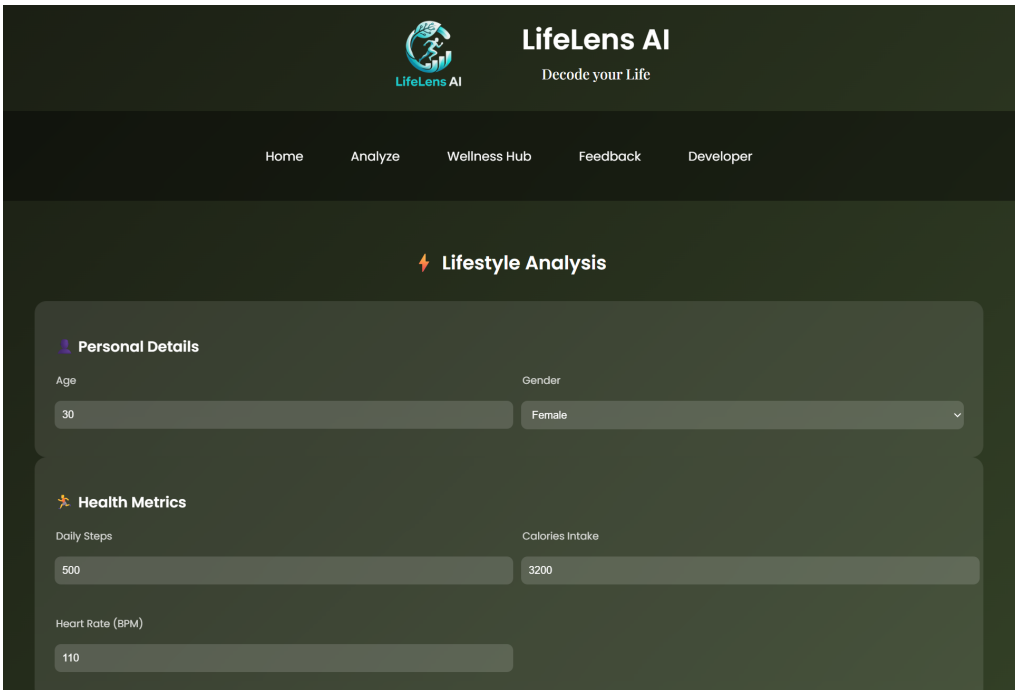


Figure 8.1: User Input Interface for Lifestyle Data Collection - Part 1

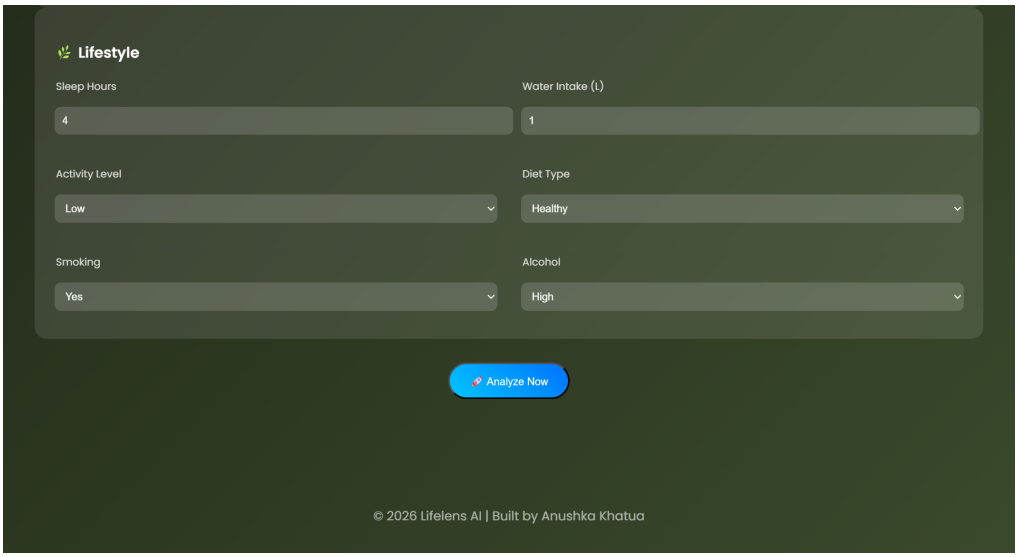


Figure 8.2: User Input Interface for Lifestyle Data Collection - Part 2

### 8.4.2 Output and Analysis Dashboard

After processing the input data, the system generates a comprehensive output dashboard displaying the results.

The dashboard includes:

- **Health Score:** Numerical representation of overall lifestyle condition
- **Risk Indicators:** Physical Health Risk, Lifestyle Risk, and Overall Risk Index
- **Key Observations:** Identification of major lifestyle concerns
- **Guidance:** Explanation of the current lifestyle status
- **Suggestions:** Personalized recommendations
- **Action Plan:** Step-by-step improvement strategies

Additionally, visual charts are provided to compare user metrics with ideal values, enhancing interpretability.

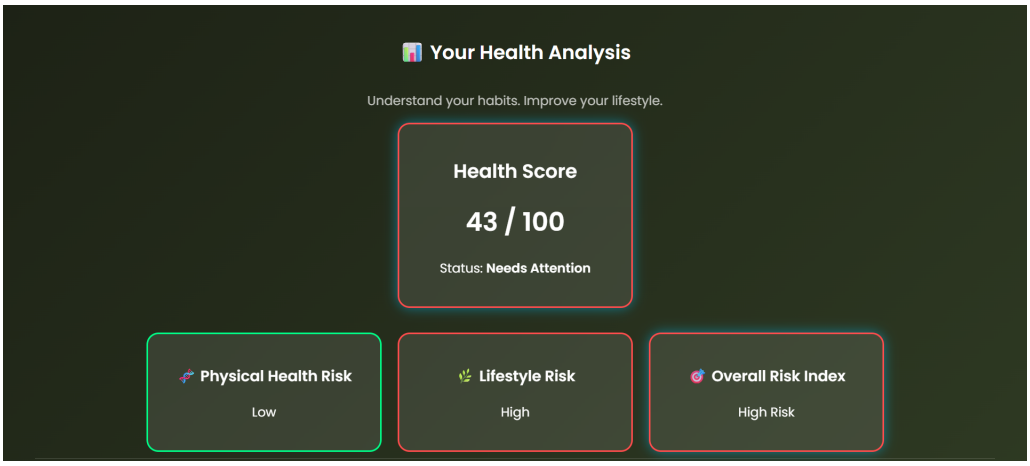


Figure 8.3: Output Dashboard Showing Risk Analysis and Insights - Part 1

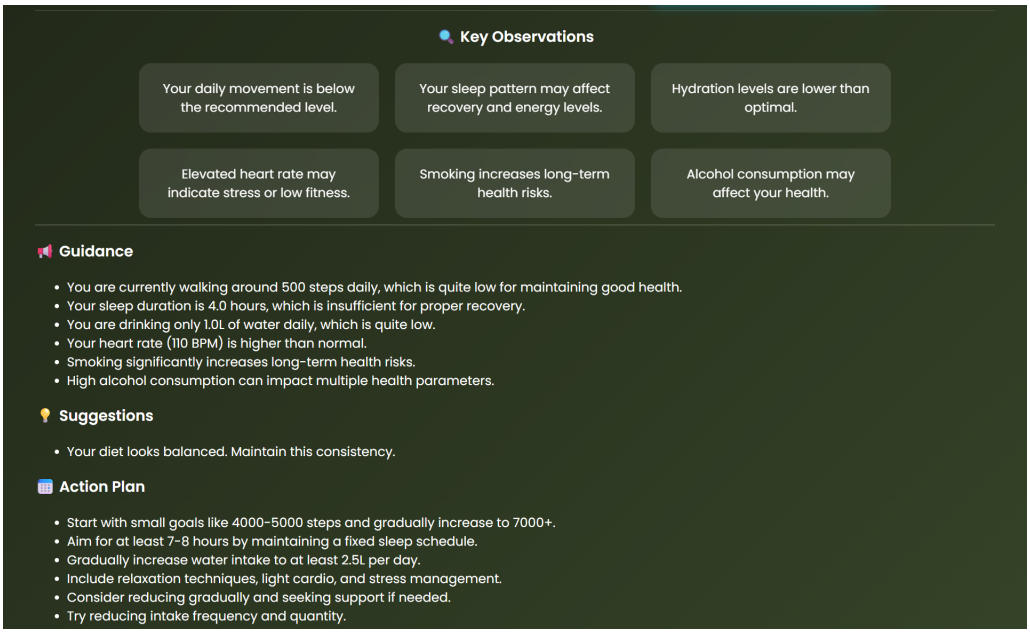


Figure 8.4: Output Dashboard Showing Risk Analysis and Insights - Part 2

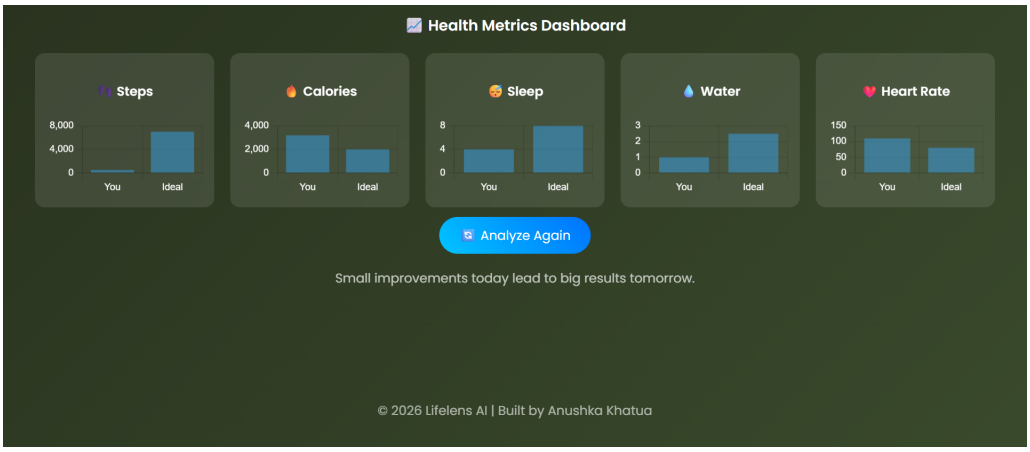


Figure 8.5: Output Dashboard Showing Risk Analysis and Insights - Part 3

### 8.4.3 System Usability

The application is designed with a modern and responsive interface, ensuring ease of use across different devices. The integration of machine learning predictions with LLM-generated explanations enhances user understanding and engagement.

The system effectively bridges the gap between technical outputs and user comprehension by presenting results in a simple and meaningful manner.

## 8.5 Model Integration

The trained machine learning and deep learning models are integrated into the backend using API endpoints. The workflow is as follows:

1. User inputs data through the frontend interface
2. Data is sent to the backend API
3. Preprocessing and feature transformation are performed
4. The ML/DL model generates a risk prediction
5. The output is passed to the LLM for explanation generation
6. Final results are displayed on the frontend dashboard

This ensures seamless interaction between system components.

## 8.6 Deployment Platform

The application is deployed using platforms such as Hugging Face Spaces, making it accessible through a web browser without requiring local installation. The deployment supports real-time interaction and ensures scalability.

## 8.7 Summary

The successful implementation and deployment of the system demonstrate its practical applicability. The integration of multiple technologies enables efficient data processing, accurate prediction, and meaningful insight generation, making the system a valuable tool for lifestyle awareness and preventive healthcare.

# Chapter 9

# Conclusion and Future Scope

## 9.1 Conclusion

This project presented an AI and LLM-powered web-based system for personalized lifestyle risk awareness. The system analyzes user lifestyle behavior data, including activity patterns, health indicators, and daily habits, to assess overall lifestyle risk levels. Machine learning and deep learning models were implemented to classify users into different risk categories, enabling accurate and data-driven risk prediction.

The integration of a Large Language Model (LLM) enhances the system by transforming model outputs into clear, human-readable insights and actionable recommendations. This significantly improves interpretability and user engagement, making the system more practical and accessible for non-technical users.

The application was successfully implemented and deployed as an interactive web platform, allowing real-time user input, prediction, visualization, and feedback. The system effectively combines predictive analytics with explainable AI, demonstrating how advanced technologies can be utilized to promote lifestyle awareness, early risk identification, and preventive healthcare practices.

## 9.2 Future Scope

The proposed system can be further improved and extended in several ways to enhance its functionality and real-world impact:

- Integration with wearable devices (such as fitness trackers and smartwatches) for continuous and real-time data collection
- Implementation of advanced deep learning architectures, such as sequence-based models, for improved temporal pattern analysis
- Development of a mobile application to increase accessibility and user engagement
- Incorporation of personalized recommendation systems using user history and behavioral trends
- Expansion to support multi-user analytics, population-level insights, and comparative analysis
- Integration with cloud-based storage and scalable deployment for handling larger datasets and users

These enhancements can significantly improve the system's scalability, accuracy, usability, and real-world applicability, making it a more comprehensive solution for lifestyle monitoring and health risk awareness.

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